Recursive 3-D Phase Unwrapping for Reliable Deformation Anomaly Detection in Multitemporal SAR Interferometry

Fengming Hu, *Member, IEEE*, Siyu Cheng, *Graduate Student Member, IEEE*, Yikai Liu, and Feng Xu, *Senior Member, IEEE*

***Abstract*— Current synthetic aperture radar (SAR) missions with short repeat times bring the opportunity to get large-scale deformations in near real-time. Since nonstationary deformation is more likely to be a risk, postclassification is widely used to identify the temporal deformation patterns within extensive interferometric SAR (InSAR) data. However, conventional 3-D phase unwrapping (PU) with the assumption of stationary deformation behavior has a high probability of unwrapping error over a long time series, leading to more false detected anomalies. Here, we proposed a recursive 3-D PU method to unwrap the 3-D data stack recursively and detect the deformation anomalies concerning the nonstationary deformation. This method involves recursive temporal PU with a temporal smoothness constraint, followed by iterative spatial PU. Then, the multiple hypothesis test (MHT) is used to determine the optimal deformation model. Scatterers with deformation anomalies are identified using a generalized ratio test and assessed by their detectability power based on the predicted phase residuals. The main advantage of the proposed algorithm is the capability of dynamic data processing and decreasing the false alarm in detected deformation anomalies. The experimental results by both the simulated and real data demonstrate that the proposed method achieves reliable 3-D PU concerning nonstationary deformation, which would be beneficial for near real-time evaluation of deformation risks.**

***Index Terms*— 3-D phase unwrapping (PU), deformation anomaly detection, dynamic process, multitemporal synthetic aperture radar (SAR) interferometry, SAR.**

I. Introduction

MULTITEMPORAL interferometric synthetic aperture radar (MT-InSAR) is an effective tool to map large-scale deformation with a millimeter-level precision. This technique overcomes the impact of the main error sources in differential InSAR, such as atmospheric delay [1] and spatio-temporal decorrelation noise [2]. The first implementation of MT-InSAR is persistent scatterer (PS) interferometry, which identifies the point scatterers showing high coherence over the whole time series [3]. Since both the atmospheric delay and deformation are spatially correlated signals, a prior deformation model is required to decrease the phase gradient [4], [5]. Then, the 3-D data stack, satisfied the assumption that the phase difference between neighboring pixels is less than π [6], can be unwrapped spatially. These PS are related to man-made infrastructures, which are mostly distributed in urban areas [7]. Considering the rural area, distributed scatterers suffering the temporal decorrelation are common. A phase reconstruction approach is required to suppress the decorrelation noise by using all combinations of interferograms [8]. Small baseline subset (SBAS) is a typical algorithm combining both PS and DS, which utilizes short temporal and spatial baseline interferograms to satisfy the PC assumption and avoid the temporal decorrelation noise [9]. With the spatially unwrapped phase, a prior deformation model is required to connect the isolated sets of interferogram clusters [10].

As the time series lengthens, the use of a linear deformation model cannot describe the complex deformation trend, leading to additional unwrapping errors. Typical kinematic models, such as seasonal periodic, temperature-related, Heaviside step, and other functions can be used to parameterize temporal characteristics of deformation [11], [12], [13]. In addition, real nonstationary deformation can be rather complex, incorporating either second-order polynomial or trigonometric function into the process would improve the estimation [14]. In [15], an M-estimation-based method is used for InSAR nonlinear deformation modeling, considering the zero-mean second-order stochastic differential equations and periodic variations. Since the deformation pattern varies with the locations, a hypothesis testing scheme is required to determine the most suitable model, avoiding the over-fitting [16]. The use of a complex deformation model will improve the accuracy of phase unwrapping (PU) but significantly extend the dimension of the estimated parameters, leading to a huge computing burden of the temporal PU. If contextual information is available for a typical geological area, it is possible to quantitatively predict the deformation and give a reasonable interpretation by using the geological model [17], [18].

In the practical application, most studies identify the spatial anomalies, e.g., scatterers whose deformation pattern is significantly different than its neighbors [19], to find the potential risks. Thus, identifying the deformation pattern is an essential step to facilitate anomaly detection, but the manual analysis is impractical due to the huge number of measurement points.

Received 9 August 2024; revised 20 January 2025; accepted 13 March 2025. Date of publication 18 March 2025; date of current version 31 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201158 and in part by Shanghai Collaborative Innovation Project under Grant XTCX-KJ-2024-13. *(Corresponding author: Fengming Hu.)*

Fengming Hu, Siyu Cheng, and Feng Xu are with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai 200433, China (e-mail: fm_hu@fudan.edu.cn; fengxu@fudan.edu.cn).

Yikai Liu is with Shanghai High School, Shanghai 200231, China.

Digital Object Identifier 10.1109/TGRS.2025.3552491

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

Postclassification can identify different types of nonstationary deformation trends within extensive InSAR data, enhancing data interpretation for nonexpert end-users. Based on the temporal displacement behavior of the scatterers, a PCA-based K-means algorithm is proposed for automatic spatial clustering of the temporal deformation patterns [20]. In [21], a limited vision source separation algorithm is adopted to automatically identify different deformation patterns in buildings and structures. The unsupervised classification gives an automated way of selecting an optimal parameterization of the InSAR deformation data for each scatterer [22].

With the emerging deep-learning methods, artificial intelligence (AI) techniques are proven to be a powerful tool considering the spatial and temporal analysis tasks in remote sensing. Typical deep-learning models, such as recurrent neural networks (RNNs) enable automatic feature extraction from high-dimensional data, which is promising to understand different deformation patterns in InSAR time series [23]. In [24], a synthetic aperture radar (SAR)-Transformer-based method is proposed to decompose InSAR time-series signals into various physics-related components, such as seasonal thermal expansion, subsidence trends, and tidal loading impacts. In addition, the deep learning-based method shows the capability of removing atmospheric noise [25] and reconstructing the noise-free postseismic deformation signals from the original InSAR data [26].

Apart from the steady-state deformation, temporal anomalies related to the nonstationary deformation are more likely to be a potential risk. Considering the changes in the deformation model, a multiple linear regression model with multiple breaks was adopted to detect the anomalous trend [27]. In addition, a long short-term memory network is proven to be efficient to handle sequential data, and detect anomalous deformation changes in InSAR time series [28], [29]. Since the reliability of the postprocess approach depends on the initial unwrapped phase, an inappropriate prior deformation model will induce additional unwrapping errors, leading to more falsely detected anomalies. A feasible approach for reliable deformation anomaly detection is testing whether the new observation fits the prevailing deformation model in a dynamic process. For example, in [30], the risk assessment of the bridge is evaluated by comparing the new observation with the established displacement model.

With a longer acquisition, the scatterers may maintain high coherence only in parts of the time series due to the change of the electromagnetic properties, which leads to reflectivity change in the time series. Such significant surface change will induce the loss of coherence, leading to phase decorrelation. Previous work identifies the step-times of the significant surface change using the amplitude time series [31]. Since the speckle noise may lead to the uncertainty of the initially detected step-times, the hybrid 3-D PU enables the refinement by the spatial–temporal consistency [32]. The proposed amplitude-augmented recursive MT-InSAR indicates that the amplitude is sensitive to the surface change since the phase can be coincidentally correct when the decorelation lead to a phase close to the expected (wrapped) value. Therefore, the use of amplitude contributes to decrease the false detected anomalies since the abnormal phase can be due to either significant surface change or deformation anomaly [33]. However, this method neglects the impact of the nonstationary deformation on the initial unwrapped phase, which has a high probability of false alarms.

Current SAR mission with a shorter repeat time and wider swath enables the opportunity for near real-time deformation monitoring but increases the computing burden. It is time-consuming to update the current results by conventional 3-D PU due to its nonlinear time complexity. An SBAS-based processing chain combining both parallel and sequential steps is developed to achieve an efficient updating [34]. The following research incorporates the M-estimation into the process to mitigate the effect of anomalous errors [35]. Assuming a temporal smooth constraint, a recursive estimator is proposed to unwrap the double differential phase measurements [36]. With the unwrapped 3-D data stack, deformation time series can be obtained using the unscented Kalman filter (UKF) [37]. In [38], a sequential phase-linking approach is introduced to get the optimal phase time series of DS. The recursive MT-InSAR achieves the recursive parameter updating and anomaly detection in the same framework [33]. However, the current dynamic process only focuses on updating the parameters of the prior deformation model, a recursive 3-D PU is required for reliable estimation of nonstationary deformation.

Here, we proposed a recursive 3-D PU for reliable deformation anomaly detection. First, the initial image stack is processed using static 3-D PU and the parameters of the prior deformation model are estimated. Then, a recursive temporal PU is proposed with the assumption of temporal smoothness constraint and followed by an iterative spatial PU. With the initial unwrapped phase, the multiple hypothesis test (MHT) is used to determine the optimal deformation model. Moreover, the generalized ratio test is utilized to confirm whether there is a significant deformation anomaly. Finally, we apply our method to both simulated and real SAR data. The results of the recursive approach are compared with those of the conventional static way.

The main contributions of this work are listed as follows.

1) A recursive 3-D PU is proposed to decrease the unwrapping errors due to the nonstationary deformation.
2) The optimal deformation model is determined to decrease the false alarms in the detected deformation anomalies.
3) The performance of the proposed method is evaluated using both simulated and real SAR data, showing its capability in near real-time deformation interpretation.

This article is organized as follows. We briefly introduce the static 3-D PU process in Section II, followed by the recursive 3-D PU, adaptive deformation model, and anomaly detection in Section II-D. Both simulated and real data results are presented in Section III, followed by the conclusion in Section IV.

II. Methodology

The flowchart of the proposed method is shown in Fig. 1, which includes three key steps, initilization, recursive 3-D PU and deformation anomaly detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

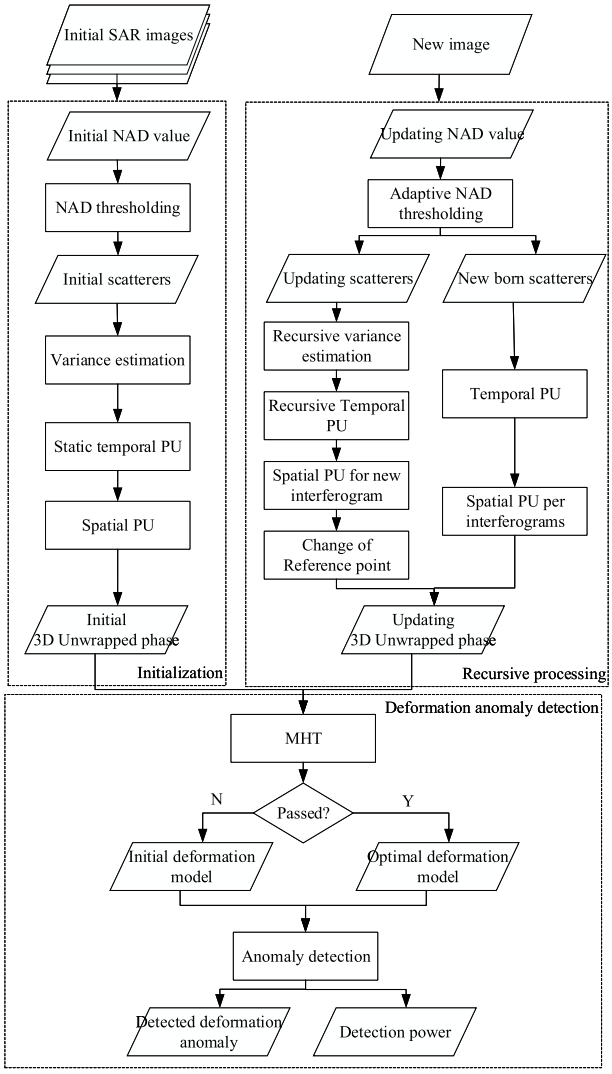


Fig. 1. Flowchart of the proposed recursive 3-D PU.

A. Initialization

Supposing that there are m co-registered SAR acquisitions, the radiometric calibration is first conducted to eliminate the difference in radiometric sensor stability. Considering the preselection of the coherent scatterers, the relative radiometric calibration is widely used due to its simple implement [13]. Then, the normalized amplitude dispersion (NAD) is computed to initially select the coherent scatterers candidates based on the calibrated amplitude time series [3]. Then, $m-1$ single primary interferograms are generated and the flat Earth effect is removed.

Since typical spatial correlated phase terms, such as deformation and atmospheric delay, will lead to multiple cycle discontinuities, conventional 3-D PU unwraps the phase difference first in the baseline domain followed by a spatial unwrapping. Thus, the fundamental measurement in MT-InSAR is the differential interferometric phases between two nearby pixels, referred to as an arc, which is defined as follows:

$$E\left\{\begin{bmatrix}\Delta\phi_{i,j}^{1}\\ \vdots\\ \Delta\phi_{i,j}^{m-1}\end{bmatrix}\right\}=\begin{bmatrix}2\pi & 0 & 0\\ 0 & \ddots & 0\\ 0 & 0 & 2\pi\end{bmatrix}\begin{bmatrix}n_{i,j}^{1}\\ \vdots\\ n_{i,j}^{m-1}\end{bmatrix}$$

$$-\frac{4\pi}{\lambda}\begin{bmatrix}\frac{B_{\perp,i}^{1}}{R_i\sin\theta_i} & B_{t_1}\\ \vdots & \vdots\\ \frac{B_{\perp,i}^{m-1}}{R_i\sin\theta_i} & B_{t_{m-1}}\end{bmatrix}\begin{bmatrix}\Delta h_{i,j}\\ \Delta v_{i,j}\end{bmatrix}+C_{i,j} \tag{1}$$

where $\Delta h_{i,j}$ and $\Delta v_{i,j}$ represent the height residual difference and the velocity difference between the two scatterers. The integer phase ambiguity is denoted by $n_{i,j}^{k}\in\mathbb{Z}$. Temporal baseline B_{t_k} and perpendicular baseline $B_{\perp,i}^{k}$ are defined alongside slant range R_i, local incidence angle θ_i, and radar wavelength λ. Phase constant $C_{i,j}$ encompasses both noise and atmospheric delay in the primary image. The operator $E\{\cdot\}$ signifies the expectation operation. The arcs were solved by using the integer least squares (ILSs) [39].

Since the reliability of the 3-D PU depends on the accuracy of the temporally unwrapped arcs, the quality of the arcs can be assessed using typical indicators, such as solution consistency and ensemble coherence. With all arc solutions, spatial PU can be applied to get the unwrapped phase of all scatterers relative to a specified reference point. Note that the coherent scatterer candidates selected by different NAD thresholds exhibit varying noise levels. Two consecutive network solutions are used to ensure both precision and densification. Details can be found in [31]. With the unwrapped phase, the parameters of interest can be estimated using a least squares approach.

Several parameters in the initialization are used as input for the subsequent recursive process.

1) *NAD Values:* Both the average and standard deviation (SD) of the whole amplitude image are retained.
2) *Mean Variance:* The estimated mean variances of the initial interferograms are retained.
3) *Coordinate:* The image coordinates of the resolved scatterers in the current epoch are retained.
4) *Unwrapped Phase:* The unwrapped phases of the resolved scatterers relative to a specified point are retained.
5) *Parameters:* The estimated parameters of the resolved scatterers, as well as their corresponding variances, are retained.

B. Recursive 3-D PU

With the initialization, the new acquisition is relatively calibrated using the stable scatterers. Supposing that the retained average and SD of the initial m amplitude images are $\overline{a}_m$ and σ_m, respectively. Then, the average and SD of the amplitude images at the $(m+1)$th epoch are updated as follows [33]:

$$\overline{a}_{m+1}=\frac{\overline{a}_m\cdot m+a_{m+1}}{m+1} \tag{2}$$

and

$$\sigma_{a_{m+1}}=\sqrt{\frac{m-1}{m}\sigma_{a_m}^{2}+\frac{(a_{m+1}-\overline{a}_m)^2}{m+1}}. \tag{3}$$

Then, the NAD image can be updated correspondingly.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

By setting an NAD threshold, the preselection will suppress the loss of the coherence caused by the significant surface change. Since the reliability of the estimated NAD depends on the length of the time series, using a fixed threshold for varying number of acquisitions will lead to fewer candidates, which increases the omission errors of potential deformation risk. As described in [32], the relative optimal NAD threshold can be obtained using the empirical probability density function of the phase SD. Supposing that the NAD threshold of the initial acquisitions is $D_{A,m}$, the quantile of the phase SD is calculated with the given level of the significance α, defined as follows:

$$T_\alpha = F^{-1}\left(1-\alpha, \sigma_\varphi | D_{A,m}; m\right) \tag{4}$$

where the cumulative density function $F(\sigma_\varphi | D_{A,m}; m)$ can be obtained by the numerical simulation.

Setting a search interval around the initial NAD threshold, we repeat the calculation and get the corresponding probabilities. In the end, the final optimal relative NAD threshold is located at the minimal difference between the estimated probability and the given significant level, defined as follows:

$$\hat{D}_{A,m+1} = \arg\min\left\{\int_0^{T_\alpha} f\left(\sigma_\varphi | D_{A,m+1}; m+1\right) \mathrm{d}\sigma_\varphi - (1-\alpha)\right\}. \tag{5}$$

Furthermore, new candidates are obtained using the relative optimal NAD thresholding method, which can be categorized into two subsets, i.e., updating candidates and new-born candidates. Updating candidates are those that intersect with the initial candidates, while the remaining new candidates are considered new-born candidates.

In the case of the updating candidate, the temporal PU of the new measurement is conducted jointly with the initial parameters. Supposing that the new observation fits the prevailing model, the predicted double differential observation of two scatters i and j at epoch m is defined as follows:

$$\Delta\hat{\phi}_{i,j}^{m|m-1} = A_m \Delta\hat{x}_{i,j}^{(m-1)} \tag{6}$$

where A_m is the design matrix that transforms the parameters into the double difference phases at mth epoch. The parameter vector $\Delta\hat{x}_{i,j}^{(m-1)}$ contains velocity and height residual differences. The notation $(m-1)$ denotes the parameters from the first to the $(m-1)$th epoch. The corresponding predicted residual is calculated as follows:

$$\Delta\hat{\phi}_{i,j,\mathrm{res}}^{m|m-1} = \mathcal{W}\left\{\Delta\varphi_{i,j}^{m} - \Delta\hat{\phi}_{i,j}^{m|m-1}\right\} \tag{7}$$

where $\mathcal{W}\{\cdot\}$ denotes the phase wrapping operator.

Assuming that the phase change caused by both the system noise and the nonstationary deformation is smaller than a quarter of the wavelength, denoted as the temporal smoothness constraint, the temporal unwrapped phase of the arc can be obtained as follows:

$$\Delta\hat{\phi}_{i,j}^{m} = \Delta\hat{\phi}_{i,j,\mathrm{res}}^{m|m-1} + \Delta\hat{\phi}_{i,j}^{m|m-1}. \tag{8}$$

With all arc solutions, an iterative spatial unwrapping is used to detect the unwrapping errors. Supposing that the reference point is the same as the initialization, the initial unwrapped phase of all scatterers at mth epoch can be obtained using the following L^1 norm solution [33]:

$$\hat{\boldsymbol{\phi}}^m = \arg\min \|\mathbf{B}\cdot\boldsymbol{\phi}^m - \Delta\hat{\boldsymbol{\phi}}^m\|^1 \tag{9}$$

where $\mathbf{B}$ signifies the design matrix that characterizes the linkage between the arcs and the scatterers. Note that the arc with a longer distance is more likely to fail the temporal smoothing constraint. Therefore, the temporal unwrapped phase may include unwrapping errors caused by large nonstationary deformation. Since scatterers in the network have a minimal number of connections, the unwrapping errors can be detected by using the spatio-temporal consistency (STC) condition, which is defined as follows [32]:

$$\Delta\hat{\boldsymbol{\phi}}_{\mathrm{STC}} = \Delta\hat{\boldsymbol{\phi}}_{\mathrm{res}} - \mathrm{round}\left\{\Delta\hat{\boldsymbol{\phi}}_{\mathrm{res}}/2\pi\right\}\cdot 2\pi \tag{10}$$

where round$\{\cdot\}$ indicates the rounding operator. Then, arcs unsatisfied with the STC condition will be removed in the subsequent iteration.

Note that the removal of the arcs will discard the potential coherent scatterers due to the poor connection. Additional arcs will be generated by linking every isolated scatterer to its nearest neighbors. These new arcs will be temporally unwrapped and checked by the STC condition. Considering the efficiency, the number of nearest points is increased sequentially and the scatterers that are connected to others more than two times will be accepted. This iteration continues until the number of the scatterer is stable.

Considering the newborn candidates, conventional static 3-D PU will be used to get the unwrapped phase. Although the number of new-born candidates is much smaller than that of the updating candidates, incorporating them into the resolved candidates will enhance the connections of the unwrapping network, especially for areas suffering from surface changes.

C. Updating Initialization

The main steps of the recursive process are prediction and updating. The prediction is conducted to detect whether the new observation satisfies the model in the initialization. If no anomaly exists, the prevailing model is sustained and the parameters in the intialization are updated using a static Kalman filter, which is defined as follows [40]:

$$\hat{x}^{(m)} = \hat{x}^{(m-1)} + G\left(\hat{\phi}^m - A_m\hat{x}^{(m-1)}\right) \tag{11}$$

with the corresponding variance

$$Q_{\hat{x}^{(m)}} = Q_{\hat{x}^{(m-1)}} - GA_m Q_{\hat{x}^{(m-1)}} \tag{12}$$

where G is the gain matrix, defined as

$$G = Q_{\hat{x}^{(m-1)}} A_m \left(Q_{\hat{\phi}^m} + A_m Q_{\hat{x}^{(m-1)}}^{-1} A_m^{\mathrm{T}}\right)^{-1} \tag{13}$$

where $Q_{\hat{x}^{(m-1)}}$ denotes the precision of the estimated parameters within the initial $m-1$ acquisitions, which is defined as follows:

$$Q_{\hat{x}^{(m-1)}} = \left(A_{m-1}^{\mathrm{T}} Q_{\hat{\phi}^{m-1}}^{-1} A_{m-1}\right)^{-1} \tag{14}$$

where $Q_{\hat{\phi}^{m-1}}$ denotes VC matrix of the $m-1$ interferograms estimated using variance component estimation (VCE) [41].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

Note that $Q_{\hat{\phi}^m}$ in (13), denoted as the variance of the new mth interferogram, is still unknown. An approximated process is given. First of all, the variance of the double-difference phase residual at mth epoch in (7) is defined as follows:

$$Q_{\Delta\hat{\phi}_{\text{res}}^{m|m-1}} = Q_{\Delta\hat{\phi}^m} + A_m Q_{\Delta\hat{x}^{(m-1)}} A_m^{\text{T}}. \tag{15}$$

Supposing that the scatterers are with the same noise level, $Q_{\Delta\hat{\phi}_{\text{res}}^{m|m-1}}$ can be approximated using the independent arcs, defined as follows:

$$Q_{\Delta\hat{\phi}_{\text{res}}^{m|m-1}} = \frac{\sum_{k=1}^{L}\left(\Delta\hat{\phi}_{k,\text{res}}^{m|m-1}\right)^2}{L} \tag{16}$$

where L denotes the number of independent arcs. Thus, the variance of the new mth interferogran is estimated as follows:

$$Q_{\hat{\phi}^m} = \frac{1}{2}\left(Q_{\Delta\hat{\phi}_{\text{res}}^{m|m-1}} - A_m Q_{\Delta\hat{x}^{(m-1)}} A_m^{\text{T}}\right). \tag{17}$$

Since the selected arcs will affect the result, the estimation of the variance component is conducted several times until the estimation is stabilized.

If the initial reference point suffers from a surface change, a new reference point should be selected. Then, the unwrapped phase of all scatterers in the initialization should be corrected as follows:

$$\hat{\boldsymbol{\phi}}_{\text{Corrected}}^{m-1} = \hat{\boldsymbol{\phi}}^{m-1} - \hat{\boldsymbol{\phi}}_{\text{new}} \tag{18}$$

where $\hat{\boldsymbol{\phi}}_{\text{new}}$ is the unwrapped phase of the new reference point. The corresponding parameters are corrected as follows:

$$\hat{x}_{\text{Corrected}}^{(m-1)} = \hat{x}^{(m-1)} - \Delta\hat{x}_{\text{ref}} \tag{19}$$

where $\Delta\hat{x}_{\text{ref}}$ denotes the difference of the parameters between the initial and the new reference point, which is calculated as follows:

$$\Delta\hat{x}_{\text{ref}} = \arg\min \|\hat{\boldsymbol{\phi}}_{\text{new}} - A_{m-1}\Delta x_{\text{ref}}\|^1. \tag{20}$$

D. Deformation Anomaly Detection

Based on the recursive 3-D PU, it is possible to detect the deformation anomalies recursively. However, since a linear deformation model is applied during the PU, the presence of potential nonstationary deformation increases the likelihood of false alarms. Thus, the optimal deformation model is determined to ensure reliable anomaly detection.

First, the overall model test (OMT) is applied to evaluate the steady-state motion [13]. A stationary deformation is determined if the test is passed. Otherwise, the nonstationary deformation exists in the unwrapped phase and the MHT should be conducted to determine the optimal deformation model. The null hypothesis H_0 assumes that the unwrapped phase follows the initial linear deformation model, while all possible alternative hypotheses H_j are derived based on a set of kinematic functions. The corresponding test statistical value is defined as follows [16]:

$$\text{T}_q^j = \hat{e}_{(m)}{}^{\text{T}} Q_{\hat{\phi}^m}^{-1} C_j \left(C_j^{\text{T}} Q_{\hat{\phi}^m}^{-1} Q_{\hat{e}_{(m)}} Q_{\hat{\phi}^m}^{-1} C_j\right)^{-1} C_j^{\text{T}} Q_{\hat{\phi}^m}^{-1} \hat{e}_{(m)} \tag{21}$$

with a χ^2 distribution with a q degrees of freedom. $\hat{e}_{(m)}$ is the phase residuals of the m acquisitions, defined as follows:

$$\hat{e}_{(m)} = \hat{\phi}^m - A_m \hat{x}^{(m)} \tag{22}$$

where $Q_{e^{(m)}}$ denotes the variances of the phase residual, defined as follows:

$$Q_{\hat{e}_{(m)}} = Q_{\hat{\phi}^m} + A_m \left(A_m^{\text{T}} Q_{\hat{\phi}^m}^{-1} A_m\right)^{-1} A_m^{\text{T}}. \tag{23}$$

C_j denotes the specification matrix related to the alternative hypothesis. Note that different alternative hypotheses have varying dimensions, leading to different χ^2 distributions. Therefore, the test statistical value is divided by its critical value and the maximal value of the ratios denotes the most probable alternative hypothesis H_B

$$T_q^B = \max_j \left\{ T_q^j / \chi_\alpha^2(q_j) \right\}. \tag{24}$$

The corresponding alternative hypothesis with the maximal statistical value is selected as the most likelihood optimal deformation model.

Furthermore, each scatterer may constains into several subsets, where the deformation is assumed to be steady-state in the temporal subsets. Note that the last subset is important in the risk assessment, whose parameters are estimated as follows:

$$\hat{x}^{(p:m)} = \left(A_{(p:m)}^{\text{T}} Q_{\hat{\phi}^{(p:m)}}^{-1} A_{(p:m)}\right)^{-1} A_{(p:m)}^{\text{T}} Q_{\hat{\phi}^{(p:m)}}^{-1} \hat{\phi}^{(p:m)} \tag{25}$$

where p denotes the step-time of the deformation trend. The notation $(p{:}m)$ means the variables are from $(p+1)$th to mth epoch.

Moreover, the deformation anomalies can be detected automatically using the following test statistical value:

$$T_1 = \frac{\left(\hat{e}_i^m\right)^2}{Q_{\hat{e}_i^m}} \tag{26}$$

with a χ^2 distribution with one degree of freedom. $\hat{e}_i^m$ denotes the phase residual, defined as follows:

$$\hat{e}_i^m = \hat{\phi}_i^{(p:m)} - A_m \hat{x}_i^{(p:m)}. \tag{27}$$

The new mth interferogram contains a deformation anomaly if $T_1 > \chi_\alpha^2(1,0)$.

Considering the alternative hypothesis, the quality metric for detected deformation anomaly can be obtained. An important value is the minimal detectable deformation (MDD) [42], which indicates whether a deformation anomaly could be detected with a detectability power γ_0 [42]. From the perspective of the users, the detectability power can be used to assess the detected deformation anomalies. Given a prior MDD, the noncentrality parameter ν_0 is obtained as

$$\nu_0 = \left(\frac{4\pi}{\lambda}\right)^2 \nabla_{\text{MDD}} Q_{\phi_i^m}^{-1} Q_{\hat{e}_i^m} Q_{\phi_i^m}^{-1} \nabla_{\text{MDD}} \tag{28}$$

and the detectability power γ_0 is found to be

$$\gamma_0^m = F\left(\chi_\alpha^2(1,0) | 1, \nu_0\right) \tag{29}$$

where $F(\cdot)$ denotes the cdf of the noncentral χ^2 distribution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

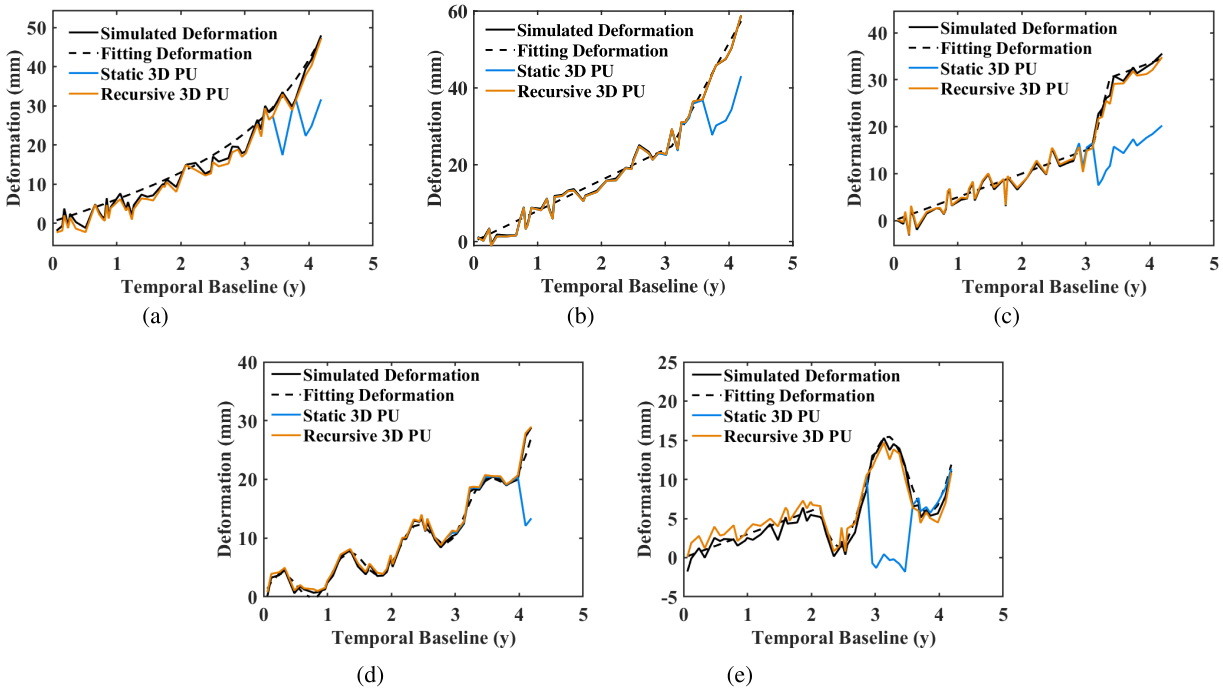


Fig. 2. Comparison of the deformation between static and recursive 3-D PU. (a)–(e) Different types of nonlinear deformation.

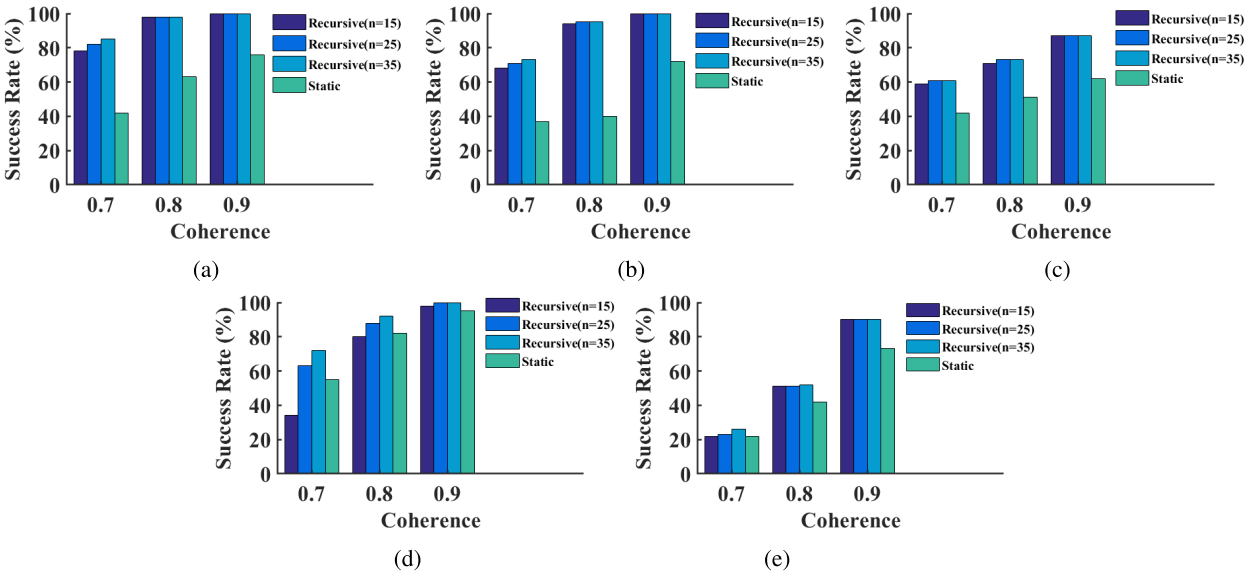


Fig. 3. Success rates of PU with varying coherence. (a)–(e) Different types of nonlinear deformation.

III. Results and Discussions

A. Case Study by Simulated Data

In this section, simulated data is utilized to verify the effectiveness of the proposed recursive 3-D PU method. The simulated dataset consists of 51 acquisitions over a period of four years, with a radar wavelength of 31.1 mm. Since this demonstration focuses on the impact of nonstationary deformation, other error sources, such as atmospheric delay and orbit error, are neglected. The height error of the scatterer is set to 30 m with the corresponding spatial baselines ranging from 10 to 600 m. Five typical deformation models, i.e., exponential, breakpoint, Heaviside periodic and piecewise periodic functions, are employed for the comparison, which is shown in Fig. 2.

The static 3-D PU assumes a linear deformation model throughout the entire time series, which can result in unwrapped errors and lead to unpredictable deformation interpretations if nonstationary deformation exists. In the case of the exponential deformation in Fig. 2(a), there is a velocity increment. The static 3-D PU incorrectly interprets this velocity increment as a time series with a similar velocity but with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

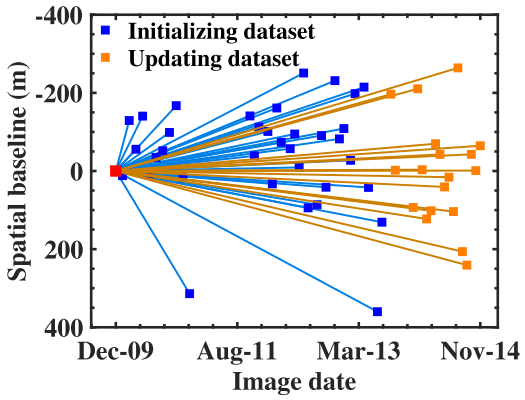


Fig. 4. Distribution of the temporal and spatial baselines. The red dot indicates the primary image; the blue dots indicate the secondary images in the initial dataset; and the yellow dots are the secondary images used in the recursive process.

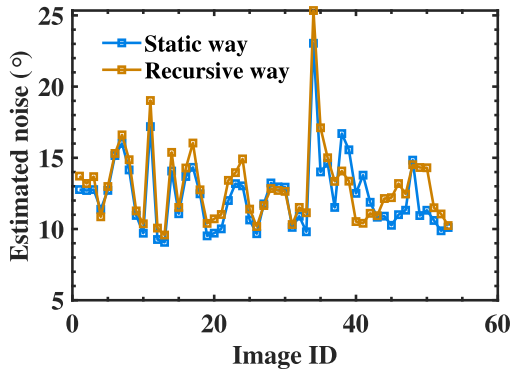
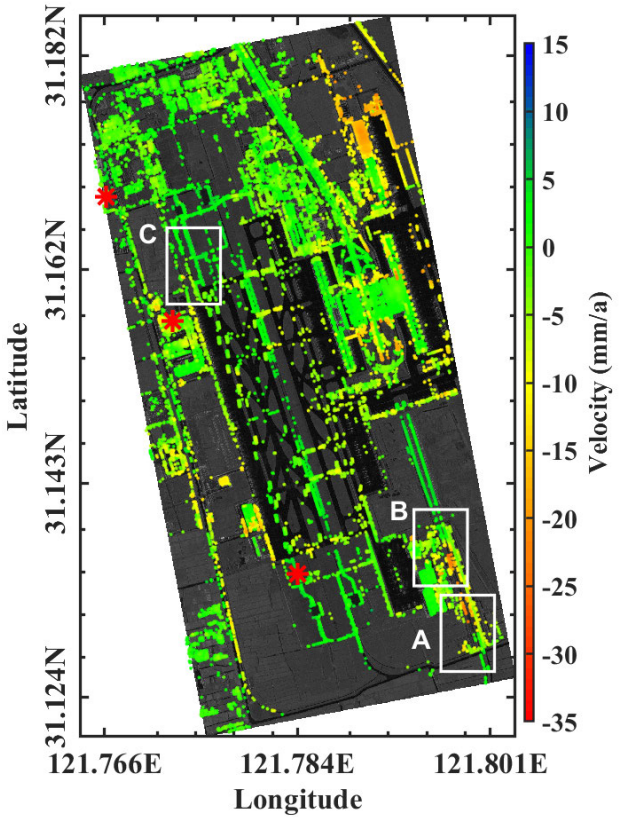


Fig. 5. Comparison of the estimated phase noise between static and recursive 3-D PU.

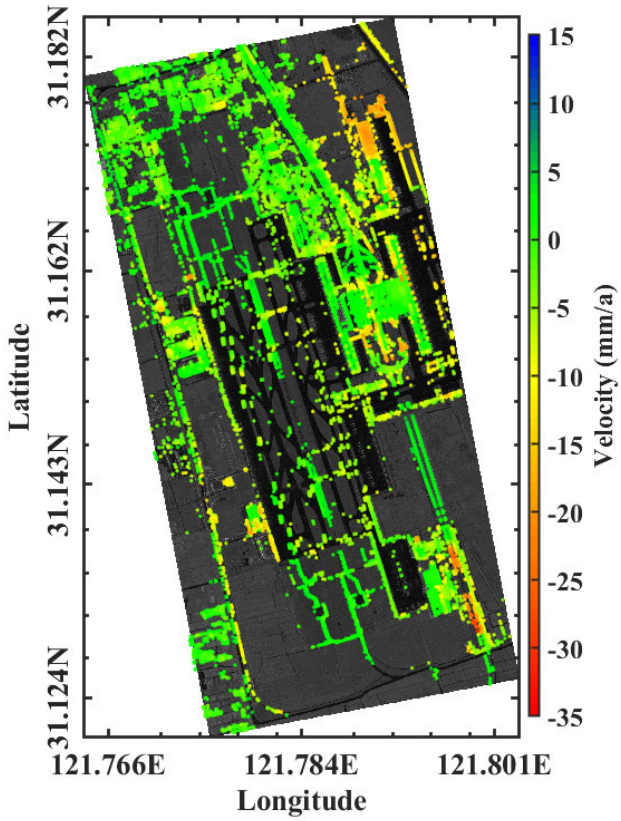
more variations. In the case of the breakpoint deformation in in Fig. 2(b), the velocity estimated by the static 3-D PU is much lower than the actual value. Although a phase jump may be detected as a deformation anomaly, the detected step-time of the anomaly occurs later than the epoch of the actual velocity change. Similar problems exist in the demonstrations based on both periodic and piecewise periodic deformation, as shown in Fig. 2(d) and (e). Considering the Heaviside function, the step-time of the deformation anomaly can be identified, which is identified as an outlier, see Fig. 2(c).

Since the optimal deformation model determined by the MHT depends on the unwrapped phase, the unwrapping errors will lead to a false model selection. Among all the demonstrations, the Heaviside deformation has the least impact on the reliability of the deformation interpretation. Other types of nonlinear deformation will bias not only the estimated velocity but also the step-time of the detected anomalies, which will increase the omission rate in the risk assessment.

The second demonstration is further testing the performance of the PU. The simulation is repeated 1000 times and the success rates of the PU are obtained, correspondingly. Note that a time series is considered successfully unwrapped only if all unwrapped phases are correct. Fig. 3 shows the comparison



(a)



(b)

Fig. 6. Comparison of the velocity between (a) static 3-D PU and (b) recursive 3-D PU.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

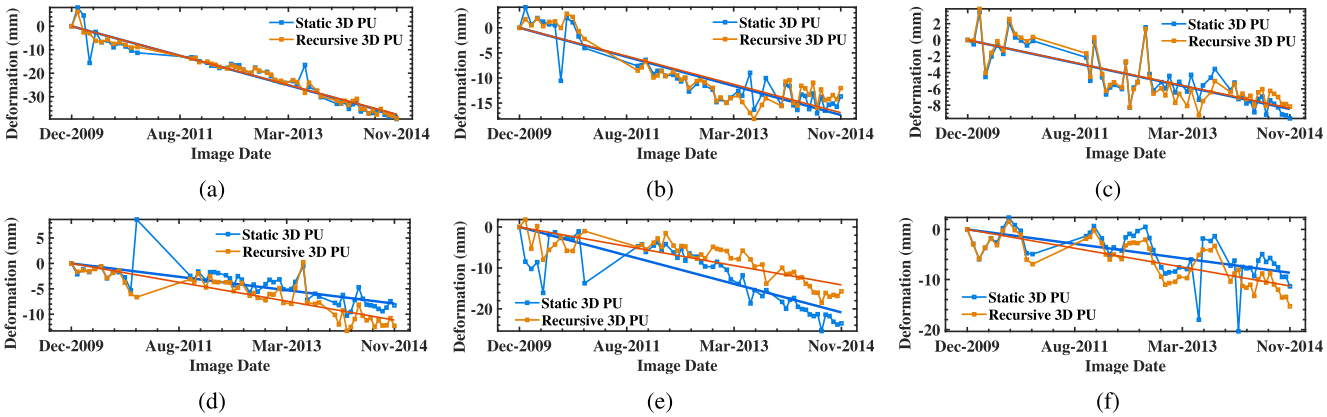


Fig. 7. Comparison of deformation time series between static and recursive 3-D PU. (a)–(c) Time series with a similar velocity. (d)–(f) Time series with a different velocity.

of the success rate between the static and the recursive 3-D PU with varying coherence. It denotes that different types of nonstationary deformation will lead to unwrapping errors in the static 3-D PU while the proposed recursive 3-D PU enables a better performance by using a relaxed temporal constraint. The demonstration of the linear deformation model in Fig. 3(a)–(c) indicates that the recursive 3-D PU significantly improves the success rate of the PU with the respect to the nonlinear deformation, especially for the low coherence configuration. Considering different types of deformation anomalies, the Heaviside function is the most challenging one due to its poor temporal continuity. Even if the coherence reaches 0.9, the success rate of the Heaviside function is 90% while that of the other functions is almost 100%.

Since the seasonal deformation is crucial for man-made structures, the second demonstration includes the periodic and the piecewise periodic functions, as shown in Fig. 3(d) and (e). If the whole time series contains the same periodic deformation, the static 3-D PU has a good performance in PU by adding the periodic function into the prior deformation model. Moreover, with the decrease of the coherence, the performance of recursive 3-D PU with a short length of the initial time series significantly degrades. If the type of the deformation is unknown, it is recommended to use more acquisitions in the initialization. While a longer initial time series can reduce the probability of unwrapping errors, it also increases the risk of including nonstationary deformation. Thus, it is important to get a tradeoff in the practical application.

B. Case Study by the Real SAR Data

This demonstration was implemented using 54 TerraSAR-X (TSX) images (December 2009–November 2014). The dataset covered a 3 × 6 km area of Shanghai Pudong International Airport. The dataset was co-registered using the GAMMA software. The slant range and azimuth pixel spacing are 0.91 and 1.71 m, respectively. During the process, no external digital elevation model (DEM) is used, the initial heights of all scatterers are set to zero to calculate the flatten phase. Then, the residual topographic phase of each scatterer relative to a specified reference scatterer is estimated using the unwrapped phase. Fig. 4 shows the distribution of the spatial and temporal

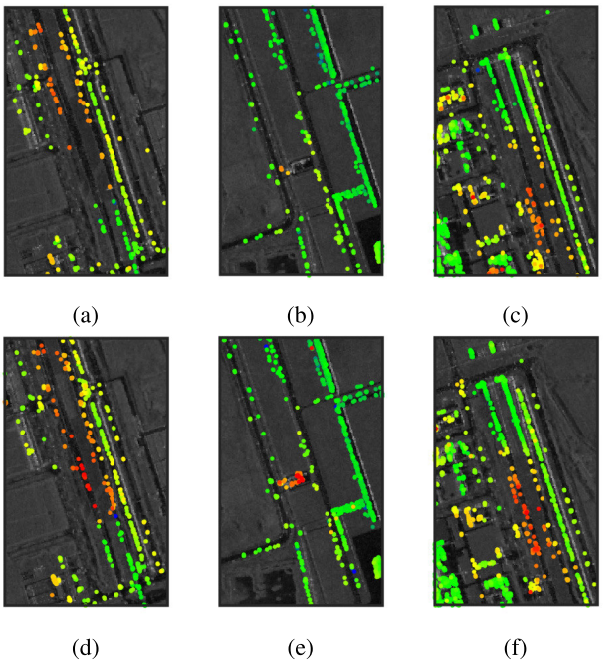


Fig. 8. Comparison of velocity in the selected areas. (a)–(c) From static 3-D PU. (d)–(f) From recursive 3-D PU. (a) and (d) Region A. (b) and (e) Region B. (c) and (f) Region C.

baseline. The maximum temporal baseline was 1500 days, and the maximum spatial baseline was 400 m. The initial stack includes the first 36 images, while the recursive stacking includes the other 18 images.

During the recursive process, both anomaly detection and parameter updating depend on the variance of the newly acquired image. Fig. 5 denotes a comparison of the estimated phase noise between the static and recursive 3-D PU. The variance estimation in the static 3-D PU was conducted using VCE [41] while that in the recursive 3-D PU was obtained using the proposed method in (15)–(17). The comparison shows a good agreement between the two approaches. The slight difference in the estimation is due to the changes in the selected scatterers.

Then, both static and recursive 3-D PU were conducted to get the velocities of the identified scatterers. Fig. 6 shows

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

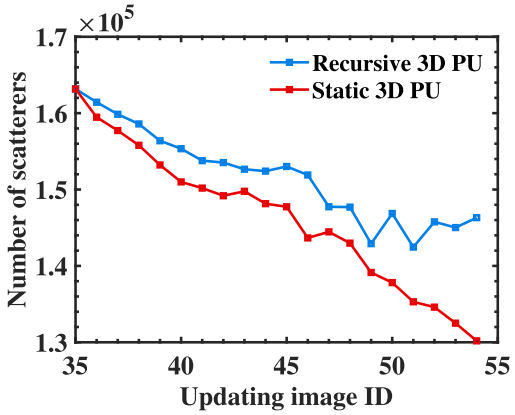


Fig. 9. Comparison of the number of scatterers between recursive and static 3-D PU.

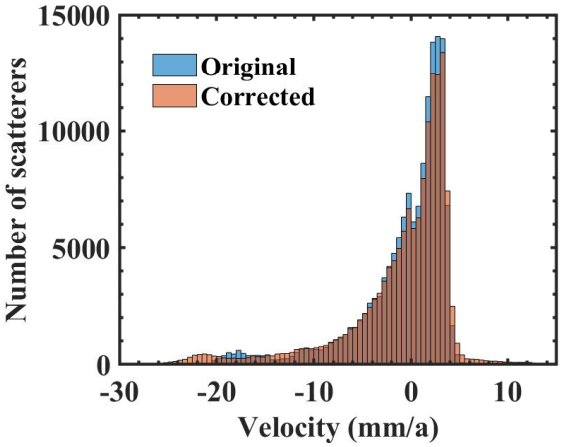


Fig. 11. Histogram of the velocities without and with correction.

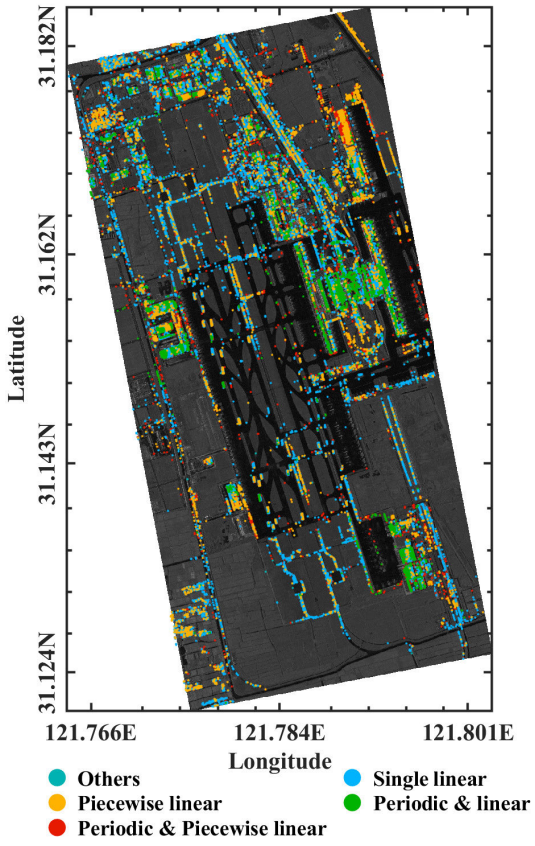


Fig. 10. Classification of the optimal deformation model obtained by the recursive 3-D PU.

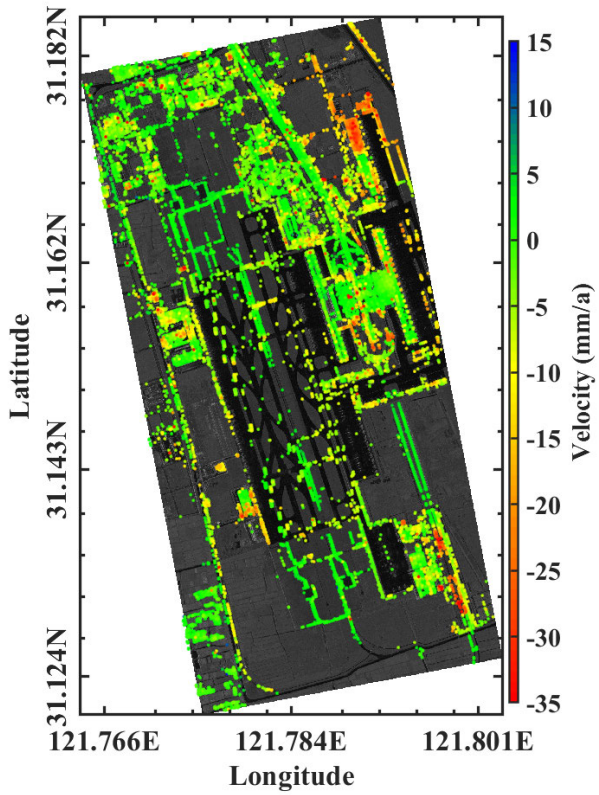


Fig. 12. Velocity map obtained by recursive 3-D PU with velocity correction.

the corresponding velocity maps, with velocities ranging from −35 to 15 mm/a. The difference in the estimated velocities of 73% of the common scatterers between the static and recursive 3-D PU is less than 0.5 mm/a, there are some scatterers with significantly different velocities due to the unwrapping errors. Six scatterers were selected to show the difference of the unwrapped phase by both static and recursive 3-D PU, as shown in Fig. 7. Fig. 7(a)–(c) denotes the scatterers with similar velocities. However, the assumption of a static linear deformation model will lead to additional deformation outliers. Moreover, extreme outliers will bias the estimated velocities, resulting in a higher rate of false anomaly detections, as shown in Fig. 7(d)–(f). The proposed recursive 3-D PU effectively eliminates these outliers.

In addition, the number of scatterers obtained by the static 3-D PU is 130k while that by the recursive 3-D PU is 146k. This difference can be attributed to two primary factors. The first factor is the inappropriate NAD threshold during the preselection of the coherent scatterers. Since the approximated bias of this indicator varies with the number of acquisitions, applying the same threshold to more acquisitions will reduce the number of candidates. Second, as the time series lengthens, a single prior deformation model is not appropriate for the whole time series, inducing more variations in the unwrapped

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

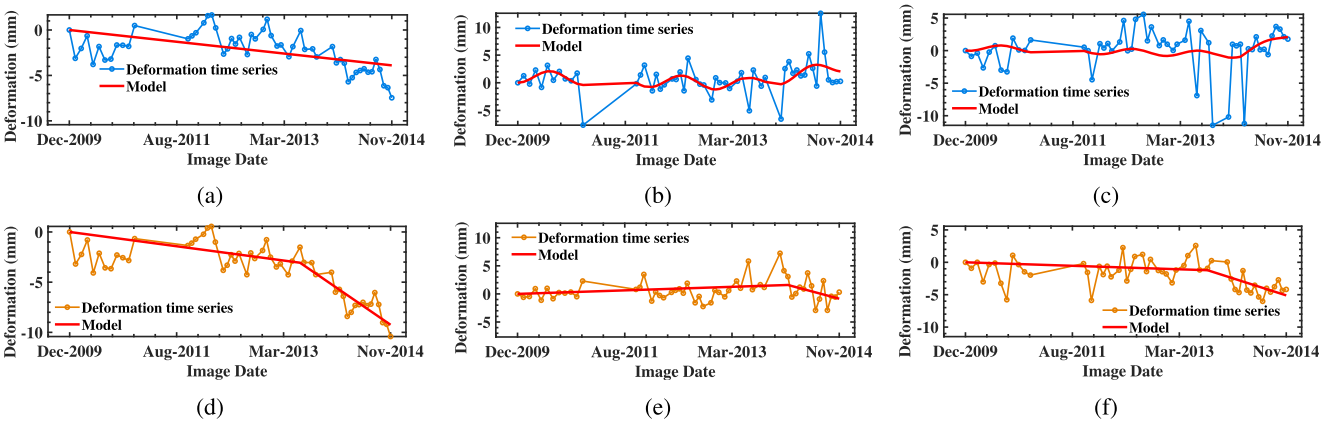


Fig. 13. Deformation time series and the adaptive model fitting of the selected scatterers. Each column denotes the result of the same scatterer. (a)–(c) Obtained by the static 3-D PU while. (d)–(f) Obtained by the recursive 3-D PU.

phase. The loss of temporal coherence will further reduce the number of coherent scatterers, which may lead to a higher probability of omission error in the risk assessment.

Three small regions are selected to show more details, which are marked as white rectangles in Fig. 6(a). The magnified views in Fig. 8 show that scatterers with significant velocity are missing in the results obtained by the static 3-D PU. As shown in Fig. 2(a) and (b), the unwrapping errors caused by the change of deformation trend result in a lower temporal coherence, so these scatterers will be discarded in the spatial PU.

In the case of the static 3-D PU, the number of identified coherent scatterers decreases with more acquisitions, as shown in Fig. 9. The loss of the scatterers will increase the difficulty in the network generation. On the contrary, the proposed recursive 3-D PU has a relatively stable number of scatterers with the accumulation of the image, which contributes to the identification of potential deformation risk.

In the practical application, the velocity map is a key product for the users to assess the risk. However, since the assumption of the stationary deformation trend often fails in a long time series, it is unsuitable to evaluate the risk using a single velocity. Based on the unwrapped phase, the MHT approach can identify the step-time of different deformation trends and get the optimal deformation models of all temporal subsets. Fig. 10 denotes a classification with the assumption of four typical deformation models. Note that a scatterer that does not belong to any one of the prior deformation models will be marked as others. Table I shows the number of scatterers within different types of deformation. It shows that more than half of the scatterers have a linear deformation behavior. The percentage of the linear deformation obtained by the static 3-D PU is 58.3% while that by the recursive 3-D PU is 55.4%. Therefore, the use of the linear deformation model results in more scatterers being misjudged as the linear deformation. These scatterers are classified as the piecewise linear deformation behavior by using the proposed recursive 3-D PU.

With the identified breakpoints in the unwrapped phase, the whole time series is divided into several subsets. Then, the original velocity is replaced by the velocity of the final subset, e.g., from the last breakpoint to the last epoch. Fig. 11 denotes the comparison of the histogram without and with the velocity correction, showing more significant velocities. Compared Fig. 12 with Fig. 6(b), the corrected velocity map is more significant than the original one, which is beneficial to the risk assessment.

In addition, unwrapping errors in the initial unwrapped will affect the optimal deformation model of the MHT, which induces more false detected anomalies. Fig. 13 shows the comparison of the deformation time series and the corresponding optimal deformation model between the static and recursive 3-D PU, indicating that the unwrapping errors will lead to a significantly different deformation model. Compared Fig. 13(a) with (d), the deformation time series by the static 3-D PU shows a single linear deformation model with a

TABLE I
NUMBER OF SCATTERERS WITHIN DIFFERENT TYPES OF DEFORMATION MODELS

Type	Static 3D PU	Recursive 3D PU
Single linear	75841	80989
Piecewise linear	5922	9541
Periodic & linear	12860	12238
Periodic & piecewise linear	33150	40086
Others	2414	3461

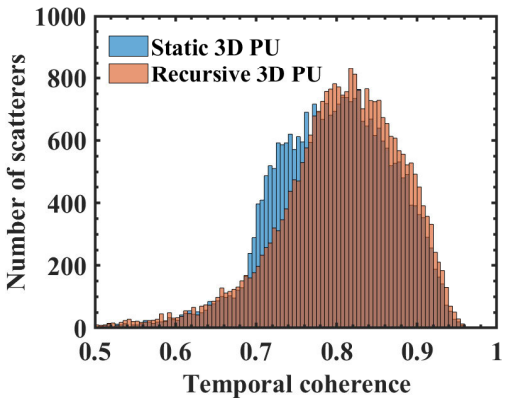


Fig. 14. Comparison of temporal coherence between static and recursive3D PU.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

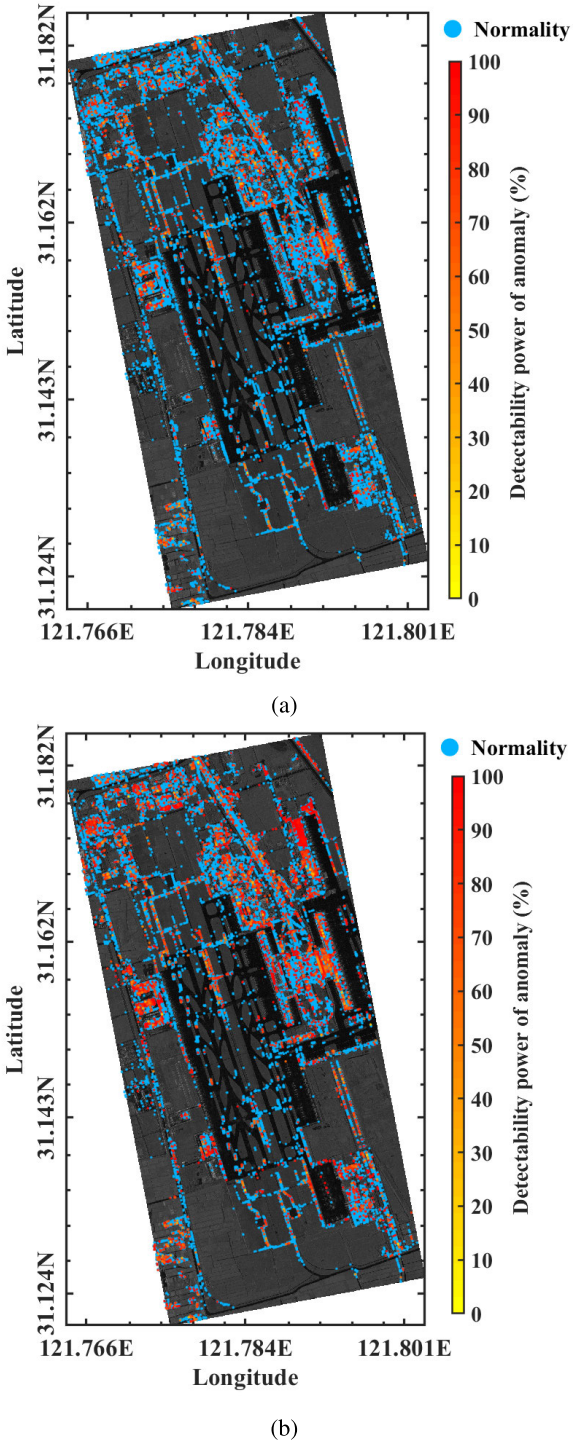


Fig. 15. Comparison of the deformation anomaly maps between (a) static and (b) recursive 3-D PU with an MDD of 10 mm.

velocity of −0.79 mm/a while that by the recursive 3-D PU is fit by a piecewise linear model. The initial velocity −0.87 mm/a and then changes to −5.00 mm/a. According to Fig. 13(b) with (c), the optimal deformation model cannot fit the deformation time series by the static 3-D PU even if both periodic and piecewise linear models are used. After eliminating the unwrapping errors by the recursive 3-D PU, the refined deformation time series can be well fit using a piecewise linear model, see Fig. 13(e) with (f). This improvement in the optimal deformation model will lead to an increase in the temporal coherence, as shown in Fig. 14.

With the identified optimal deformation model, a χ^2-test is utilized to detect the deformation anomalies with a level of significance of 0.02. From the perspective of the users, the reliability of the detected deformation anomalies is necessary in the practical application. A useful approach is evaluating whether a certain displacement is a potential risk using the detectability power [33]. Setting the MDD to 10 mm, the corresponding deformation anomaly maps by both static and recursive 3-D PU are shown in Fig. 15. Since the reliability of the prior deformation model has a significant impact on the detected anomalies, the proposed recursive 3-D PU decreases the probability of false alarms in the detected anomalies.

IV. Conclusion

In this article, a recursive 3-D PU is proposed for the reliable detection of deformation anomalies concerning nonstationary deformation. Assuming a temporal smoothness constraint, an initial unwrapped arc is obtained using temporal PU. Subsequently, an iterative spatial PU is applied to remove noise and correct unwrapping errors based on the STC condition. In addition, the MHT is utilized to determine the optimal deformation, and the deformation anomalies are obtained by the generalized ratio test. Experimental results by both simulated and real SAR data show that the existing unwrapping errors will significantly change the types of the identified deformation model and bias the interpretation of the deformation. The proposed recursive 3-D PU enables fewer unwrapping errors due to the nonstationary deformation, which will not only improve the identification of the deformation patterns but also decrease the false alarms in the detected deformation anomalies. Thus, this research would contribute to the automatic interpretation of InSAR deformation for near real-time risk assessment.

Acknowledgment

The authors acknowledge the German Aerospace Center DLR for the scientific use of TerraSAR-X images.

References

[1] H. A. Zebker, P. A. Rosen, and S. Hensley, "Atmospheric effects in interferometric synthetic aperture radar surface deformation and topographic maps," *J. Geophys. Res., Solid Earth*, vol. 102, no. B4, pp. 7547–7563, Apr. 1997.

[2] R. F. Hanssen, *Radar Interferometry: Data Interpretation and Error Analysis*. Dordrecht, The Netherlands: Kluwer, 2001.

[3] A. Ferretti, C. Prati, and F. Rocca, "Permanent scatterers in SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 1, pp. 8–20, Apr. 2001.

[4] A. Ferretti, C. Prati, and F. Rocca, "Nonlinear subsidence rate estimation using permanent scatterers in differential SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 5, pp. 2202–2212, Sep. 2000.

[5] B. M. Kampes and R. F. Hanssen, "Ambiguity resolution for permanent scatterer interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 11, pp. 2446–2453, Nov. 2004. [Online]. Available: http://doris.tudelft.nl/Literature/kampes04.pdf

[6] D. C. Ghiglia and M. D. Pritt, *Two-Dimensional Phase Unwrapping: Theory, Algorithms, and Software*. Hoboken, NJ, USA: Wiley, 1998.

[7] J. Wu and F. Hu, "Monitoring ground subsidence along the Shanghai Maglev zone using TerraSAR-X images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 117–121, Jan. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.

[8] G. Fornaro, S. Verde, D. Reale, and A. Pauciullo, "CAESAR: An approach based on covariance matrix decomposition to improve multibaseline–multitemporal interferometric SAR processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2050–2065, Apr. 2015.

[9] P. Berardino, G. Fornaro, R. Lanari, and E. Sansosti, "A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 11, pp. 2375–2383, Nov. 2002.

[10] O. Mora, J. J. Mallorqui, and A. Broquetas, "Linear and nonlinear terrain deformation maps from a reduced set of interferometric SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 10, pp. 2243–2253, Oct. 2003.

[11] O. Monserrat, M. Crosetto, M. Cuevas, and B. Crippa, "The thermal expansion component of persistent scatterer interferometry observations," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 864–868, Sep. 2011.

[12] F. Hu, F. J. V. Leijen, L. Chang, J. Wu, and R. F. Hanssen, "Monitoring deformation along railway systems combining multi-temporal InSAR and LiDAR data," *Remote Sens.*, vol. 11, no. 19, p. 2298, Oct. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/19/2298

[13] F. J. van Leijen, "Persistent scatterer interferometry based on geodetic estimation theory," Ph.D. dissertation, Dept. Geosci. Remote Sens., Delft Univ. Technol., Delft, The Netherlands, 2014.

[14] R. Zhang et al., "An integrated model for extracting surface deformation components by PSI time series," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 2, pp. 544–548, Feb. 2014.

[15] Y. Su et al., "An M-estimation method for InSAR nonlinear deformation modeling and inversion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5207912.

[16] L. Chang and R. F. Hanssen, "A probabilistic approach for InSAR time-series postprocessing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 421–430, Jan. 2016.

[17] W. Xiang et al., "Modeling saline mudflat and aquifer deformation synthesizing environmental and hydrogeological factors using time-series InSAR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11134–11147, 2021.

[18] X. Xing et al., "Quantitative prediction for deformation and brine extraction in salt solution mining based on water-solution kinetic (WSK) InSAR model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7655–7673, 2024.

[19] F. Cigna and D. Tapete, "Present-day land subsidence rates, surface faulting hazard and risk in Mexico city with 2014–2020 Sentinel-1 IW InSAR," *Remote Sens. Environ.*, vol. 253, Feb. 2021, Art. no. 112161. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425720305344

[20] D. Festa et al., "Unsupervised detection of InSAR time series patterns based on PCA and K-means clustering," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Mar. 2023, Art. no. 103276. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1569843223000985

[21] G. Martin, A. Hooper, T. J. Wright, and S. Selvakumaran, "Blind source separation for MT-InSAR analysis with structural health monitoring applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7605–7618, 2022.

[22] B. van de Kerkhof, V. Pankratius, L. Chang, R. van Swol, and R. F. Hanssen, "Individual scatterer model learning for satellite interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1273–1280, Feb. 2020.

[23] Z. Wu, P. Ma, Y. Zheng, F. L. Gu, L. Liu, and H. Lin, "Automatic detection and classification of land subsidence in deltaic metropolitan areas using distributed scatterer InSAR and oriented R-CNN," *Remote Sens. Environ.*, vol. 290, Mar. 2023, Art. no. 113545. [Online]. Available: https://api.semanticscholar.org/CorpusID:257787837

[24] P. Ma, Z. Wu, Z. Zhang, and F. T. K. Au, "SAR-transformer-based decomposition and geophysical interpretation of InSAR time-series deformations for the Hong Kong-Zhuhai-Macao bridge," *Remote Sens. Environ.*, vol. 302, Mar. 2024, Art. no. 113962. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S003442572300514X

[25] Z. Zhao, Z. Wu, Y. Zheng, and P. Ma, "Recurrent neural networks for atmospheric noise removal from InSAR time series with missing values," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 227–237, Oct. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271621002094

[26] C. Li, X. Xi, G. Zhang, X. Song, and X. Shan, "A deep-learning neural network for postseismic deformation reconstruction from InSAR time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4505214.

[27] M. Zhu et al., "Detection of building and infrastructure instabilities by automatic spatiotemporal analysis of satellite SAR interferometry measurements," *Remote Sens.*, vol. 10, no. 11, p. 1816, Nov. 2018.

[28] A. Kulshrestha, L. Chang, and A. Stein, "Use of LSTM for sinkhole-related anomaly detection and classification of InSAR deformation time series," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4559–4570, 2022.

[29] R. Salih Kuzu et al., "Automatic detection of building displacements through unsupervised learning from InSAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6931–6947, 2023.

[30] Q. Huang, M. Crosetto, O. Monserrat, and B. Crippa, "Displacement monitoring and modelling of a high-speed railway bridge using C-band Sentinel-1 data," *ISPRS J. Photogramm. Remote Sens.*, vol. 128, pp. 204–211, Jun. 2017.

[31] F. Hu, J. Wu, L. Chang, and R. F. Hanssen, "Incorporating temporary coherent scatterers in multi-temporal InSAR using adaptive temporal subsets," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7658–7670, Oct. 2019.

[32] F. Hu, Y. Gong, S. Cheng, and F. Xu, "Amplitude-aided 3-D phase unwrapping for temporary coherent scatterers interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5217915.

[33] F. Hu, F. J. van Leijen, L. Chang, J. Wu, and R. F. Hanssen, "Combined detection of surface changes and deformation anomalies using amplitude-augmented recursive InSAR time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5210816.

[34] M. Manunta et al., "The parallel SBAS approach for Sentinel-1 interferometric wide swath deformation time-series generation: Algorithm description and products quality assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6259–6281, Sep. 2019.

[35] J. Xu, M. Jiang, V. G. Ferreira, and Z. Wu, "Time-series InSAR dynamic analysis with robust sequential adjustment," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[36] Q. Verburg, "QUInSAR: Temporal parameter and ambiguity estimation using recursive least-squares: A methodology for persistent scatterer interferometry," M.S. thesis, Dept. Geosci. Remote Sens., Delft Univ. Technol., Delft, The Netherlands, 2017.

[37] F. Xue and X. Lv, "Applying time series interferometric synthetic aperture radar and the unscented Kalman filter to predict deformations in Maoxian landslide," *J. Appl. Remote Sens.*, vol. 13, no. 1, Feb. 2019, Art. no. 014509.

[38] H. Ansari, F. De Zan, and R. Bamler, "Sequential estimator: Toward efficient InSAR time series analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5637–5652, Oct. 2017.

[39] P. J. G. Teunissen, "Integer aperture GNSS ambiguity resolution," *Artif. Satell.*, vol. 38, no. 3, pp. 79–88, 2003.

[40] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.

[41] L. Zhang, "Temporarily coherent point SAR interferometry," Ph.D. dissertation, Dept. Comput., Hong Kong Polytech. Univ., Hong Kong, 2012. [Online]. Available: https://theses.lib.polyu.edu.hk/handle/200/6737

[42] L. Chang, R. P. B. J. Dollevoet, and R. F. Hanssen, "Monitoring line-infrastructure with multisensor SAR interferometry: Products and performance assessment metrics," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1593–1605, May 2018.



Fengming Hu (Member, IEEE) received the B.E. and Ph.D. degrees from Tongji University, Shanghai, China, in 2014 and 2020, respectively.

From October 2017 to October 2019, he was a Guest Ph.D. Student at Delft University of Technology, Delft, The Netherlands. From 2020 to 2023, he was a Post-Doctoral at Fudan University, Shanghai, where he has been a Research Associate with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), since 2023. His interests include multitemporal synthetic aperture radar (SAR) interferometry, microwave vision radar 3-D imaging, and multitemporal change detection using synthetic aperture radar images.

Dr. Hu is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.



Siyu Cheng (Graduate Student Member, IEEE) received the B.E. degree in electronic information engineering from Nanjing University of Information Science and Technology, Nanjing, China, in 2023. She is currently pursuing the M.E. degree in communication engineering with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai, China.

Her research interests are dynamic data processing of interferometric synthetic aperture radar (InSAR) and its applications in geological disaster monitoring.



Yikai Liu has been a Student of Shanghai High School, Shanghai, China, since 2023, and participated in the joint Program of Fudan University and Shang High School. His research interests include mathematics and image processing.



Feng Xu (Senior Member, IEEE) received the B.E. degree (Hons.) in information engineering from Southeast University, Nanjing, China, in 2003, and the Ph.D. degree (Hons.) in electronic engineering from Fudan University, Shanghai, China, in 2008.

He is currently a Professor and the Vice Dean of the School of Information Science and Technology and the Vice Director of the Key Laboratory for Information Science of Electromagnetic Waves (MoE). His research interests include electromagnetic scattering modeling, synthetic aperture radar (SAR) information retrieval, and radar system development.

Dr. Xu was a recipient of the Second-Class National Nature Science Award of China in 2011, the 2014 Early Career Award of the IEEE Geoscience and Remote Sensing Society, and the 2007 SUMMA Graduate Fellowship in the advanced electromagnetics area. He is the Founding Chair of the IEEE GRSS Shanghai Chapter and an IEEE GRSS AdCom Member. He is a Topic Associate Editor of IEEE TRANSACTIONS OF GEOSCIENCE AND REMOTE SENSING and an Associate Editor of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2014 to 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:18 UTC from IEEE Xplore. Restrictions apply.